

Cs302 Mcqs final term quiz

Question # 1 of 10 (Start time: 03:03:55 PM) Total Marks: 1

Divide-by-32 counter can be achieved by using

Select correct option:

Flip-Flop and DIV 10

Flip-Flop and DIV 16

Flip-Flop and DIV 32

DIV 16 and DIV 32

Question # 2 of 10 (Start time: 03:05:20 PM) Total Marks: 1

The counter states or the range of numbers of a counter is determined by the formula. (“n” represents the total number of flip-flops)

Select correct option:

(n raise to power 2)

(n raise to power 2 and then minus 1)

(2 raise to power n)

(2 raise to power n and then minus 1)

Question # 3 of 10 (Start time: 03:06:36 PM) Total Marks: 1

A 4- bit UP/DOWN counter is in DOWN mode and in the 1010 state. on the next clock pulse, to what state does the counter go?

Select correct option:

1001

1011

0011

1100

Question # 4 of 10 (Start time: 03:07:37 PM) Total Marks: 1

A 4-bit binary UP/DOWN counter is in the binary state zero. the next state in the DOWN mode is _____

Select correct option:

0001

1111

1000

1110

Question # 5 of 10 (Start time: 03:09:04 PM) Total Marks: 1

Divide-by-160 counter is achieved by using

Select correct option:

Flip-Flop and DIV 10

Flip-Flop and DIV 16

DIV 16 and DIV 32

DIV 16 and DIV 10

Question # 6 of 10 (Start time: 03:10:01 PM) Total Marks: 1

A counter is implemented using three (3) flip-flops, possibly it will have _____ maximum output status.

Select correct option:

3

7

8

15

Question # 7 of 10 (Start time: 03:10:49 PM) Total Marks: 1

RCO stands for _____

Select correct option:

Reconfiguration Counter Output

Ripple Counter Output

Reconfiguration Clock Output

Ripple Clock Output

Question # 8 of 10 (Start time: 03:11:38 PM) Total Marks: 1

_____ occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

Select correct option:

Race condition

Clock Skew

Ripple Effect

None of given options

Question # 9 of 10 (Start time: 03:12:20 PM) Total Marks: 1

For a down counter that counts from (111 to 000), if current state is "101" the next state will be _____

Select correct option:

111

110

010

none of given options

Question # 10 of 10 (Start time: 03:13:03 PM) Total Marks: 1

A Divide-by-20 counter can be achieved by using

Select correct option:

Flip-Flop and DIV 10

Flip-Flop and DIV 16

Flip-Flop and DIV 32

Div 10 and DIV 16

Q : Ripple Clock Output

The 74HC163 is a 4-bit Synchronous Counter.it has.....data output pins

Select correct option:

2

4

6

8

Q : _____ Counters as the name indicates are not triggered simultaneously.

Select correct option:

Asynchronous

Synchronous

Positive-Edge triggered

Negative-Edge triggered

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Q : A counter is implemented using three (3) flip-flops, possibly it will have _____ maximum output status.

Select correct option:

3

7

8

15

Q : Design of state diagram is one of many steps used to design

Select correct option:

a clock

a truncated counter

an UP/DOWN counter

any counter

Q : A synchronous decade counter will have _____ flip-flops.

Select correct option:

3

4

7

10

Q : Karnaugh map is used in designing.

Select correct option:

a clock

a counter
an UP/DOWN counter
All of the above

Q : _____ is said to occur when multiple internal variables change due to change in one input variable

Select correct option:

Hold and Wait
Clock Skew
Race condition
Hold delay

Q : Three cascaded modulus-10 counters have an overall modulus of

Select correct option:

30
100
1000
10000

Q : An Astable multivibrator is known as a(n) _____

Select correct option:

Oscillator
Booster
One-shot
Dual-shot

Q: _____ occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

Select correct option:

Race condition
Clock Skew
Ripple Effect
None of given options

Q: The glitches due to "Race Condition" can be avoided by using a _____

Select correct option:

Gated flip-flops

Pulse triggered flip-flops

Positive-Edge triggered flip-flops

Negative-Edge triggered flip-flops

Q: In case of cascading Integrated Circuit counters, the enable inputs and RCO of the Integrated Circuit counters allow cascading of multiple counters together

Select correct option:

True

False

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Quiz: A flip-flop is presently in SET state and must remain SET on the next clock pulse. What must J and K be?

Select correct option:

$J = 1, K = 0$

$J = 1, K = X$ (Don't care)

$J = X$ (Don't care), $K = 0$

$J = 0, K = X$ (Don't care)

Q: The Synchronous counters are also known as Ripple Counters:

Select correct option:

True

False

Q: A decade counter can be implemented by truncating the counting sequence of a MOD-20 counter.

Select correct option:

True

False

Quiz: The terminal count of a 4-bit binary counter in the DOWN mode is_____

Select correct option:

0000

0011

1100

1111

Quiz: An Asynchronous Down-counter is implemented (Using J-K flip-flop) by connecting_____

Select correct option:

Q output of all flip-flops to clock input of next flip-flops

Q' output of all flip-flops to clock input of next flip-flops

Q output of all flip-flops to J input of next flip-flops

Q' output of all flip-flops to K input of next flip-flops

the terminal count of a modulus-13 binary counter is

Select correct option:

0000

1111

1101

1100

Quiz: A decade counter can be implemented by truncating the counting sequence of a MOD-20 counter.

Select correct option:

True

False

Quiz: A 4- bit UP/DOWN counter is in DOWN mode and in the 1010 state. on the next clock pulse, to what state does the counter go?

Select correct option:

1001

1011

0011

1100

Quiz: Design of state diagram is one of many steps used to design

Select correct option:

a clock

a truncated counter

an UP/DOWN counter

any counter

Quiz: An Astable multivibrator is known as a(n) _____

Select correct option:

Oscillator

Booster

One-shot

Dual-shot

Quiz: The glitches due to "Race Condition" can be avoided by using a _____

Select correct option:

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Pulse triggered flip-flops

Positive-Edge triggered flip-flops

Negative-Edge triggered flip-flops

Quiz: A decade counter is _____

Select correct option:

Mod-3 counter

Mod-5 counter
Mod-8 counter
Mod-10 counter

Question # 1 of 10 (Start time: 03:40:29 PM) Total Marks: 1

The terminal count of a 4-bit binary counter in the DOWN mode is_____

Select correct option:

0000

0011

1100

1111

Question # 2 of 10 (Start time: 03:40:50 PM) Total Marks: 1

The Synchronous counters are also known as Ripple Counters:

Select correct option:

True

False

Question # 3 of 10 (Start time: 03:41:08 PM) Total Marks: 1

_____occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

Select correct option:

Race condition

Clock Skew

Ripple Effect

None of given options

Question # 4 of 10 (Start time: 03:41:27 PM) Total Marks: 1

Divide-by-160 counter is achieved by using

Select correct option:

Flip-Flop and DIV 10

Flip-Flop and DIV 16

DIV 16 and DIV 32

DIV 16 and DIV 10

Question # 5 of 10 (Start time: 03:41:44 PM) Total Marks: 1

Design of state diagram is one of many steps used to design

Select correct option:

- a clock
- a truncated counter
- an UP/DOWN counter
- any counter**

Question # 6 of 10 (Start time: 03:42:01 PM) Total Marks: 1

In a 4-bit binary counter, the next state after the terminal count in the DOWN mode is _____

Select correct option:

- 0000
- 1111**
- 0001
- 10000

The 74HC163 is a 4-bit Synchronous Counter.it has.....data output pins

Select correct option:

- 2
- 4**
- 6
- 8

Q : _____ Counters as the name indicates are not triggered simultaneously.

Select correct option:

- Asynchronous**
- Synchronous
- Positive-Edge triggered
- Negative-Edge triggered

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Q : A counter is implemented using three (3) flip-flops, possibly it will have _____ maximum output status.

Select correct option:

3

7

8

15

Q : Design of state diagram is one of many steps used to design

Select correct option:

a clock

a truncated counter

an UP/DOWN counter

any counter

Q : A synchronous decade counter will have _____ flip-flops.

Select correct option:

3

4

7

10

Q : Karnaugh map is used in designing.

Select correct option:

a clock

a counter

an UP/DOWN counter

All of the above

Q : _____ is said to occur when multiple internal variables change due to change in one input variable

Select correct option:

Hold and Wait

Clock Skew

Race condition

Hold delay

Q : Three cascaded modulus-10 counters have an overall modulus of

Select correct option:

30

100

1000

10000

Q : An Astable multivibrator is known as a(n) _____

Select correct option:

Oscillator

Booster

One-shot

Dual-shot

Q: _____ occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

Select correct option:

Race condition

Clock Skew

Ripple Effect

None of given options

Q: The glitches due to "Race Condition" can be avoided by using a _____

Select correct option:

Gated flip-flops
Pulse triggered flip-flops
Positive-Edge triggered flip-flops
Negative-Edge triggered flip-flops

Q: In case of cascading Integrated Circuit counters, the enable inputs and RCO of the Integrated Circuit counters allow cascading of multiple counters together
Select correct option:

True
False

vuhelp.pk

Quiz: A flip-flop is presently in SET state and must remain SET on the next clock pulse. What must J and K be?
Select correct option:

$J = 1, K = 0$
 $J = 1, K = X$ (Don't care)
 $J = X$ (Don't care), $K = 0$
 $J = 0, K = X$ (Don't care)

Q: The Synchronous counters are also known as Ripple Counters:
Select correct option:

True
False

Q: A decade counter can be implemented by truncating the counting sequence of a MOD-20 counter.
Select correct option:

True
False

Quiz: The terminal count of a 4-bit binary counter in the **DOWN mode** is _____
Select correct option:

0000
0011
1100
1111

Quiz: An Asynchronous Down-counter is implemented (Using J-K flip-flop) by connecting_____

Select correct option:

Q output of all flip-flops to clock input of next flip-flops
Q' output of all flip-flops to clock input of next flip-flops
Q output of all flip-flops to J input of next flip-flops
Q' output of all flip-flops to K input of next flip-flops

the terminal count of a modulus-13 binary counter is

Select correct option:

0000
1111
1101
1100

Quiz: A decade counter can be implemented by truncating the counting sequence of a MOD-20 counter.

Select correct option:

True
False

Quiz: A 4- bit UP/DOWN counter is in DOWN mode and in the 1010 state. on the next clock pulse, to what state does the counter go?

Select correct option:

1001
1011

0011

1100

Quiz: Design of state diagram is one of many steps used to design

Select correct option:

a clock

a truncated counter

an UP/DOWN counter

any counter

Quiz: An Astable multivibrator is known as a(n) _____

Select correct option:

Oscillator

Booster

One-shot

Dual-shot

Quiz: The glitches due to "Race Condition" can be avoided by using a _____

Select correct option:

Gated flip-flops

Pulse triggered flip-flops

Positive-Edge triggered flip-flops

Negative-Edge triggered flip-flops

Quiz: A decade counter is _____

Select correct option:

Mod-3 counter

Mod-5 counter

Mod-8 counter

Mod-10 counter

Question # 1 of 10 (Start time: 03:40:29 PM) Total Marks: 1

The terminal count of a 4-bit binary counter in the DOWN mode is _____

Select correct option:

0000

0011

1100

1111

Question # 2 of 10 (Start time: 03:40:50 PM) Total Marks: 1

The Synchronous counters are also known as Ripple Counters:

Select correct option:

True

False

Question # 3 of 10 (Start time: 03:41:08 PM) Total Marks: 1

_____occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

Select correct option:

Race condition

Clock Skew

Ripple Effect

None of given options

Question # 4 of 10 (Start time: 03:41:27 PM) Total Marks: 1

Divide-by-160 counter is achieved by using

Select correct option:

Flip-Flop and DIV 10

Flip-Flop and DIV 16

DIV 16 and DIV 32

DIV 16 and DIV 10

Question # 5 of 10 (Start time: 03:41:44 PM) Total Marks: 1

Design of state diagram is one of many steps used to design

Select correct option:

a clock

a truncated counter

*an UP/DOWN counter
any counter*

Question # 6 of 10 (Start time: 03:42:01 PM) Total Marks: 1

*In a 4-bit binary counter, the next state after the terminal count in the DOWN mode
is_____*

Select correct option:

- 0000*
 - 1111***
 - 0001*
 - 10000*
-

Divide-by-160 counter is acheived by using
Select correct option:

- Flip-Flop and DIV 10
- Flip-Flop and DIV 16
- DIV 16 and DIV 32
- DIV 16 and DIV 10**

Divide-by-32 counter can be achieved by using
Select correct option:

- Flip-Flop and DIV 10
- Flip-Flop and DIV 16_
- Flip-Flop and DIV 32
- DIV 16 and DIV 32**

A Divide-by-20 counter can be acheived by using
Select correct option:

- Flip-Flop and DIV 10**
- Flip-Flop and DIV 16
- Flip-Flop and DIV 32
- Div 10 and DIV 16

**how can we achieved these counters is there any formula or method to
know about this ?**

In the keyboard encoder, how many times per second does the ring counter scan the key
board?

600 scans/second

625 scans/second

650 scans/second

700 scans/second

when the transmission line is idle in an asynchronous transmission

It is set to logic low

It is set to logic high

It remains in previous state

State of transmission line is not used to start transmission

A divide-by-10 Johnson counter requires

ten flip-flops

four flip-flops

five flip-flops

twelve flip-flops

A stage in the shift register consists of

a latch

a flip flop

a byte of storage

four bits of storage

When the number of states are reduced during the design any counter

output changes and Input remain unchanged

Input changes and output remain unchanged

Input and output bit change

Input and output remain unchanged

Number of states in an 8-bit Johnson counter sequence are

8

12

14

16

Question # 6 of 20 (Start time: 11:30:56 PM) Total Marks: 1

The bit capacity of a memory that has 1024 addresses and can store 8 bits at each address is:

Select correct option:

8

1024

4096

8192

• CS302 Digital Logic Design Quiz No.4 Fall 2013

Question # 1 of 10 (Start time: 03:03:55 PM) Total Marks: 1

Divide-by-32 counter can be achieved by using

Select correct option:

Flip-Flop and DIV 10

Flip-Flop and DIV 16 _____ = **Answer**

Flip-Flop and DIV 32

DIV 16 and DIV 32

Question # 2 of 10 (Start time: 03:05:20 PM) Total Marks: 1

The counter states or the range of numbers of a counter is determined by the formula. ("n" represents the total number of flip-flops)

Select correct option:

(n raise to power 2)

(n raise to power 2 and then minus 1)

(2 raise to power n) _____ = **Answer**

(2 raise to power n and then minus 1)

Question # 3 of 10 (Start time: 03:06:36 PM) Total Marks: 1

A 4- bit UP/DOWN counter is in DOWN mode and in the 1010 state. on the next clock pulse, to what state does the counter go?

Select correct option:

1001 _____ = **Answer**

1011

0011

1100

Question # 4 of 10 (Start time: 03:07:37 PM) Total Marks: 1

A 4-bit binary UP/DOWN counter is in the binary state zero. the next state in the DOWN mode is _____

Select correct option:

0001

1111 _____ = **Answer**

1000

1110

Question # 5 of 10 (Start time: 03:09:04 PM) Total Marks: 1
Divide-by-160 counter is achieved by using
Select correct option:

Flip-Flop and DIV 10
Flip-Flop and DIV 16
DIV 16 and DIV 32
DIV 16 and DIV 10 = Answer

Question # 6 of 10 (Start time: 03:10:01 PM) Total Marks: 1
A counter is implemented using three (3) flip-flops, possibly it will
have _____ maximum output status.
Select correct option:

3
7
8 = Answer
15

Question # 7 of 10 (Start time: 03:10:49 PM) Total Marks: 1
RCO stands for _____
Select correct option:

Reconfiguration Counter Output
Ripple Counter Output
Reconfiguration Clock Output
Ripple Clock Output = Answer

Question # 8 of 10 (Start time: 03:11:38 PM) Total Marks: 1
_____ occurs when the same clock signal arrives at different times
at different clock inputs due to propagation delay.
Select correct option:

Race condition = Answer
Clock Skew
Ripple Effect
None of given options

Question # 9 of 10 (Start time: 03:12:20 PM) Total Marks: 1
For a down counter that counts from (111 to 000), if current state is

"101" the next state will be _____

Select correct option:

111

110

010

none of given options = Answer

Question # 10 of 10 (Start time: 03:13:03 PM) Total Marks: 1

A Divide-by-20 counter can be achieved by using

Select correct option:

Flip-Flop and DIV 10 = Answer

Flip-Flop and DIV 16

Flip-Flop and DIV 32

Div 10 and DIV 16

Aug 19, 2014



• [+ M.Tariq Malik](#)

Solved CS302 Digital Logic Design Online Quiz No. 04

Question # 1 of 10 (Start time: 11:54:42 PM) Total Marks: 1

The cell marked 6 in 4-variable K-Map represent minterm 6 or the maxterm 6 having the following binary value of variables A, B, C and D.

Select correct option:

A=1, B=1, C=0, D=0

A=0, B=1, C=1, D=0 (lec 10 p98) = Correct Choice

A=0, B=0, C=1, D=1

A=1, B=0, C=0, D=1

Question # 2 of 10 (Start time: 11:56:07 PM) Total Marks: 1

A + B = B + A is _____

Select correct option:

Demorgan's Law

Distributive Law

Commutative Law = Correct Choice

Associative Law

Question # 3 of 10 (Start time: 11:56:50 PM) Total Marks: 1

The output of an XOR gate is zero (0) when _____ I) All the inputs are zero II) Any of the inputs is zero III) Any of the inputs is one IV) All the inputs are one

Select correct option:

I Only

IV Only

I and IV only = Correct Choice

II and III only

sec(s)

Question # 4 of 10 (Start time: 11:58:17 PM) Total Marks: 1

For a Standard SOP expression

, a _____ is placed in the cell corresponding to the product term

(Minterm) present in the expression.

Select correct option:

0

1 = Correct Choice

X (don't care condition)

Any of given option depending on SOP term.

sec(s)

Question # 5 of 10 (Start time: 11:59:45 PM) Total Marks: 1

NOR Gate can be used to perform the operation of AND, OR and NOT Gate

Select correct option:

TRUE

FALSE = Correct Choice

sec(s)

Question # 6 of 10 (Start time: 12:01:06 AM) Total Marks: 1

An example of SOP expression is

Select correct option:

$A + B(C + D)$

$A'B + AC' + AB'C$

$(A' + B + C)(A + B' + C)$

both (a) and (b) = Correct Choice

Question # 7 of 10 (Start time: 12:02:29 AM) Total Marks: 1

A standard POS form has _____ terms that have all the variables in the domain of the expression.

Select correct option:

Sum = Correct Choice

Product

Min

Composite

Question # 8 of 10 (Start time: 12:02:56 AM) Total Marks: 1

Don't care conditions are marked as _____ in the output column of the function table

Select correct option:

0

1

X = Correct Choice

None of the given options

Question # 9 of 10 (Start time: 12:04:15 AM) Total Marks: 1

The 4-variable Karnaugh Map (K-Map) has _____ cells for min or max terms

Select correct option:

4

8

12

16 = Correct Choice

Question # 10 of 10 (Start time: 12:05:44 AM) Total Marks: 1
High level Noise Margins (VNH) of CMOS 5 volt series circuits is

Select correct option:

0.3 V

0.5 V

0.9 V = Correct Choice

Question # 1 of 10 (Start time: 11:29:09 PM) Total Marks: 1
Bi-stable devices remain in either of their _____ states unless the inputs force the device to switch its state

Select correct option:

Ten

Eight

Three

Two

Question # 2 of 10 (Start time: 11:30:10 PM) Total Marks: 1
In Synchronous systems, the output of all the digital circuits changes when an enable signal is applied instead of the clock signal.

Select correct option:

True

False (Page No. 228)

Question # 3 of 10 (Start time: 11:30:31 PM) Total Marks: 1
_____ flip-flops are obsolete now.

Select correct option:

Edge-triggered

Master-Slave (Page No. 257)

T-Flipflop

D-Flipflop

Question # 4 of 10 (Start time: 11:31:21 PM) Total Marks: 1

If a circuit suffers "Clock Skew " problem, the output of circuit can't be guaranteed.

Select correct option:

True

False

Question # 5 of 10 (Start time: 11:31:36 PM) Total Marks: 1

Flip flops are also called _____

Select correct option:

Bi-stable multivibrators (Page No. 228)

Bi-stable single vibrators

Bi-stable dual vibrators

Bi-stable transformer

Question # 6 of 10 (Start time: 11:33:01 PM) Total Marks: 1

A mono-stable device only has a single stable state

Select correct option:

True (Page No. 262)

False

Question # 7 of 10 (Start time: 11:33:43 PM) Total Marks: 1

If $S=1$ and $R=0$, then for positive edge triggered flip-flop $Q(t+1) =$

Select correct option:

0

1 (Page No. 230)

Invalid

Input is invalid

Question # 8 of 10 (Start time: 11:34:03 PM) Total Marks: 1

If the S and R inputs of the gated S-R latch are connected together using a _____ gate then there is only a single input to the latch. The input is represented by D instead of S or R (A gated D-Latch)

Select correct option:

AND

OR

NOT (Page No. 226)

XOR

Question # 9 of 10 (Start time: 11:35:17 PM) Total Marks: 1

In Master-Slave flip-flop the Clock signal is connected to Slave flip-flop using _____ gate

Select correct option:

AND (Not Confirmed)

OR

NOT

NAND

Question # 10 of 10 (Start time: 11:36:46 PM) Total Marks: 1

The glitches due to "Race Condition" can be avoided by using a

Select correct option:

Gated flip-flops

Pulse triggered flip-flops

Positive-Edge triggered flip-flops

Negative-Edge triggered flip-flops (Page No. 267)